

Parallel Computing and Network Systems in Healthcare

1. Parallel Computing in Healthcare

How it's used

Modern medical imaging techniques, such as Magnetic Resonance imaging (MRI), Computed Tomography (CT) scans & Positron Emission Tomography (PET) scans, generate vast amounts of data require efficient processing. Parallel computing plays a vital role in this context by enabling the rapid processing of these large data sets.

1. Data Processing: Algorithms for image reconstruction and enhancement are executed on Graphics processing Units (GPUs) or multiCore processors, allowing for faster computation.

2. AI Based Diagnosis: Deep learning models, particularly Convolutional Neural Networks (CNNs) are employed for automated diagnosis. These models are trained & executed using parallel processing, which significantly speeds up the analysis of imaging data.

Why it's important

- Speed: Reduces image processing time from hours to minutes or seconds.
- Accuracy: Enables more precise and detailed images, improving diagnostic quality.
- AI Diagnosis: Supports real time AI-assisted evaluations during surgeries or emergencies.

2. Network Systems in health Care

How it's used

Networked systems are integral to the management & sharing of patient data across health care facilities. These systems facilitate the storage access & sharing of critical information, including medical images, lab results & clinical notes.

1. **Electronic Health Records (EHRs):** Hospitals & clinics utilize EHRs to maintain comprehensive patient records that can be accessed by authorized personnel.
2. **Remote Access:** Imaging data processed in one location can be accessed by health care providers in different hospital.
3. **Telemedicine:** Network systems enable specialists to collaborate remotely sharing images and reports seamlessly.
4. **cloud-based health platforms:** These platforms allow for the integration of AI diagnostic tools, making them accessible to various health care providers.

Why it's important

- **Data sharing:** Facilitates quick access to patient records for better diagnosis & treatment.
- **Remote Care:** Allows doctors in rural or underserved areas to consult with specialists.
- **Security & backup:** Ensure data protection & recovery in case of system failure.